

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), DECEMBER 2019

Course Code: PHT100

Course Name: ENGINEERING PHYSICS A
(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

- 1 List any six points to compare electrical oscillator with a mechanical oscillator. (3)
- 2 Distinguish between transverse and longitudinal waves. Give one example for each. (3)
- 3 When a medium of $\mu \neq 1$ is introduced in the Newton's ring set up, what happens to the diameter of interference pattern? Explain it with the help of relevant equation. (3)
- 4 Give 3 differences between interference and diffraction. (3)
- 5 State Heisenberg's Uncertainty principle and write the three uncertainty relations. (3)
- 6 Explain the optical properties of nanomaterials. (3)
- 7 Distinguish between magnetic induction and magnetising field. (3)
- 8 Derive the equation of continuity for time varying fields. (3)
- 9 Show that superconductors are perfect diamagnets. (3)
- 10 Distinguish between step index and graded index fibres. (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Set up the differential equation for a forced harmonic oscillator and solve it. (10)
b) A transverse wave on a stretched string is described by $y(x,t)=2\sin(20t+0.021x+\pi/6)$ where x and y are in cm and t is in second. Obtain (1)Amplitude (2)Initial phase (3)speed (4)frequency (4)
- 12 a) Derive an expression for the fundamental frequency of a transverse wave in a stretched string. (10)
b) A sitar wire is under tension of 40 N and length of the bridge is 80cm. A 10m (4)

sample of that wire has mass **1.2g**. Find the speed and fundamental frequency of transverse wave on the wire.

Module-II

- 13 a) With necessary diagram, write the formation of interference pattern in an air wedge and derive an expression for the diameter of a thin wire. (10)
- b) A monochromatic light of wavelength **5893 Å** is incident normally on a soap film of $\mu = 1.42$. What is the least thickness of the film that will appear dark by reflection ? (4)
- 14 a) Derive the grating equation and describe an experiment to determine the wavelength of light. Define resolving power of a grating with expression. (10)
- b) A grating has **6000 lines/cm**. Find angular separation between two wavelengths **577nm** and **579 nm** in the second order. (4)

Module-III

- 15 a) Derive an expression for energy eigen values and normalised wave function for a particle in a box of width L. (10)
- b) Calculate the separation between the two lowest energy levels of an electron in a one dimensional box of width **4Å** in joules. Given $m_e = 9.1 \times 10^{-31} \text{ kg}$; $h = 6.625 \times 10^{-34} \text{ Js}$ (4)
- 16 a) Write a note on quantum confinement and based on this explain nano sheets, nano wire and quantum dots. (10)
- b) Mention any four applications of nanotechnology. (4)

Module-IV

- 17 a) State Gauss' law in magnetism, Ampere's circuital law, Faraday's laws of electromagnetic induction and Lenz's law. Give their equations. (10)
- b) A magnetising field of **1800 A/m** produces a magnetic flux of **$3 \times 10^{-5} \text{ Wb}$** in an iron bar of cross-sectional area **0.2 cm^2** . Calculate the permeability. (4)
- 18 a) Starting from Maxwell's equations derive the expression for the velocity of electromagnetic waves in vacuum. (10)
- b) State and explain Poynting's theorem. (4)

Module-V

- 19 a) Explain the characteristics of Type I and Type II superconductors with appropriate diagrams and examples. (7)
- b) Discuss BCS theory of superconductivity. Give any four applications of (7)

superconductivity.

- 20 a) Explain construction and working of a solar cell and draw its I-V characteristics. Mention any two applications of solar cells. (10)
- b) The numerical aperture of an optic fibre is **0.295** and refractive index of core is **1.54**. Calculate refractive index of cladding and acceptance angle. (4)
